

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Observation Task No.: 1

Student Name: J Satya Srikar

Roll No.: A24126552259

Branch & Section: CSM-B

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1. TITLE

Develop structured web page using fundamental building blocks of HTML and HTML5

2. AIM

To understand and implement the fundamental building blocks of HTML and HTML5 for creating structured web pages.

3. OBJECTIVE

The objectives of this experiment are:

- To study the basic tags and formatting components in standard HTML templates.
 - To utilize superscript (<sup>) and subscript (<sub>) text markers for custom typography.
 - To implement semantic HTML5 page layout sections including <header>, <nav>, <section>, <article>, and <footer>.
 - To build complex tabular structures displaying student databases and placement statistics.
 - To construct comprehensive forms capturing user profiles, radio choices, checkbox groups, ranges, number lists, and color pickers.
 - To embed interactive local multimedia elements using HTML5 <audio> and <video> elements.
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4. THEORY

HTML (HyperText Markup Language) serves as the core semantic skeletal framework for modern webpages. While historical HTML relied purely on layout dividers (div blocks), HTML5 introduces descriptive tags such as <header>, <nav>, <section>, <article>, and <footer> to describe document layouts clearly for search crawlers and screen readers. Additionally, HTML5 provides built-in validation schemes for interactive forms and direct media rendering capabilities (audio and video nodes) natively without external plugin binaries.

Theory of HTML5 Document Elements:

- **Semantic Blocks:** Layout divisions are organized logically: <header> hosts page banner titles, <nav> provides links pools, <article> frames discrete stories, <section> divides databases grids, and <footer> displays closing notes.
 - **Subscripts & Superscripts:** Visual modifiers <sup> and <sub> shift text vertical offsets up or down, useful for displaying equations or notations.
 - **Tables:** Formatted grids structuring rows (<tr>), headers (<thead>), body records (<tbody>), and columns (<td>).
 - **HTML5 Form Inputs:** Collect diverse user datasets via type selectors. Validation checks are triggered for email, password masks, radio buttons (single choice), checkboxes (multiple select options), color arrays, range sliders, and numeric counters.
 - **Native Multimedia Nodes:** Media assets are embedded directly via <audio> and <video> tags using the controls attribute, enabling play, pause, and volume controls without requiring external Flash plugins.
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5. ALGORITHM / PROCEDURE

1. Set up project workspace folders: Observation/Observation task 1/Html, Docs, and Media.
 2. Create college-portal.html inside the Html folder and declare standard HTML5 headers.
 3. Add page title information and insert local anchor linkages for student portals.
 4. Implement the header component featuring subscript/superscript text offsets.
 5. Create a Flexbox navigation alignment section containing lists pointing to portal links.
 6. Incorporate an article block displaying the ANITS college logo image next to description texts.
 7. Design a placement details data table containing student roll numbers, names, and companies.
 8. Assemble an input fieldset container containing name text inputs, emails, passwords, radio selectors, checkboxes, range sliders, numbers, and dates.
 9. Embed standard audio and video nodes using controls parameters inside media dividers.
 10. Compile, execute the webpage inside Google Chrome, and verify formatting behaviors.
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6. PROGRAM

File: college-portal.html

college-portal.html - Part 1 (Lines 1-80)

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>

<body>
  <header>
    <center>
      <h1>ANITS</h1>
    </center>
    <center>
      <p>VIZAG<sup>IN</sup><sub>Near Tagarapuvalasa</sub></p>
    </center>
    <p>
      Anil Neerukonda Institute of Technology & Sciences (ANITS) is an autonomous engineering
      institution located in Sangivalasa, near Visakhapatnam, established in 2001. Affiliated with
      Andhra University and accredited with a NAAC "A+" grade, the college offers a range of B.Tech
      programs, including Computer Science and Engineering, Artificial Intelligence & Machine Learning,
      and Data Science. Spread across a 12-acre campus, ANITS combines strong academic infrastructure
      with active placement support, giving students a solid foundation to build technical skills and
      launch careers in the tech industry.
    </p>
  </header>
  <br>
  <nav>
    <div>
      <ol>
        <a href="#studentDetails">
          <li>Home</li>
        </a>
        <a href="#studentDetails">
          <li>Student Portal</li>
        </a>
        <li>Alumni</li>
        <li>About us</li>
      </ol>
      <ul>
        <a href="https://www.anits.edu.in/">
          <li>Main Portal</li>
        </a>
        <a href="">
          <li>Teacher Portal</li>
        </a>
        <li>Alumni</li>
        <li>About us</li>
      </ul>
    </div>
  </nav>
  <article>
    <div>
      
      <p>
        Anil Neerukonda Institute of Technology & Sciences (ANITS), Visakhapatnam is owned by Megha Engineering
        and
        Infrastructure Limited (MEIL) which is one of the top infrastructure and manufacturing company in India
        with
        headquarters at Hyderabad. The institute was established in the academic year 2001-02 by Anil Neerukonda
        Educational Society (ANES) which was founded by Dr. N.B.R. Prasad, an NRI Philanthropist from the USA,
```

college-portal.html - Part 2 (Lines 81-153)

```
</center>
</section>
<div class="form">
  <fieldset>
    <legend>
      The form
    </legend>
    <form action="post">
      <label for="name">Name</label>
      <input type="text" id="name" placeholder="Name"><br>
      <label for="email">Email</label>
      <input type="email" name="email" id="email" placeholder="Email"><br>
      <label for="password">Password</label>
      <input type="password" name="password" id="password"><br>

      <label>Gender</label><br>
      <input type="radio" name="gender" id="genderMale">
      <label for="genderMale">Male</label>
      <br>
      <input type="radio" name="gender" id="genderFemale">
      <label for="genderFemale">Female</label>
      <br>

      <label>Hobbies</label><br>
      <input type="checkbox" name="Hobbies" id="hobbySport">
      <label for="hobbySport">Sport</label>
      <br>
      <input type="checkbox" name="Hobbies" id="hobbySinging">
      <label for="hobbySinging">Singing</label>
      <br>
      <input type="checkbox" name="Hobbies" id="hobbyFreeFire">
      <label for="hobbyFreeFire">Free Fire</label>
      <br>

      <label for="branch">Branch</label>
      <select name="Branch" id="branch">
        <option value="CSM">CSM</option>
        <option value="CSD">CSD</option>
        <option value="CSE">CSE</option>
      </select>

      <label for="favColor">Color</label>
      <input type="color" id="favColor">

      <label for="range1">Range</label>
      <input type="range" id="range1">

      <label for="number1">Number</label>
      <input type="number" id="number1">

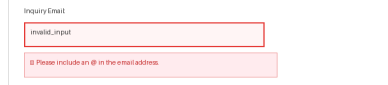
      <label for="date1">Date</label>
      <input type="date" id="date1">
    </form>
  </fieldset>
</div>
<div class="media">
  <audio src="./song.mp3" controls></audio>
  <br>
  <video src="./video.mp4" controls></video>
  <br>
  <iframe width="360" height="200" src="https://www.youtube.com/embed/LTQ_5xntgsw"
    title="YouTube video" frameborder="0">
  </iframe>
</div>
</div>
```

7. CODE EXPLANATION

HTML Tag / Code	Explanation
<sup> / <sub>	Displays text in superscript (raised vertical offset) / subscript (lowered vertical offset).
<header> / <nav>	Semantic structural sections for page introduction banners and link lists.
<article> / <section>	Defines standalone article segments / data-driven structural blocks on pages.
	Loads local image file path dynamically on screen alongside description texts.

8. TEST CASES AND EXECUTION DATA (WITH RELEVANT SCREENSHOTS)

The test suite below documents verified execution in Chrome, including 1 negative test case demonstrating an input validation failure:

TC ID	Test Description	Input / Action	Expected Result	Relevant Execution Screenshot	Status
TC-01	Text Formatting & Hierarchy	Inspect heading and paragraph tags	Document displays clear typographic hierarchy		Pass
TC-02	Anchor Navigation	Click '#academics' link	Smooth scroll navigates to targeted content section		Pass
TC-03	Data Table Layout	Inspect placement statistics table	Tabular matrix renders clean row and column borders		Pass
TC-04	Multimedia Node Playback	Trigger native audio/video controls	Media stream decodes and plays without external plugins		Pass
TC-05	Email Pattern Validation	Submit invalid 'student.email'	Submission halts; browser flags missing '@' in email field		Fail

9. OUTPUT

Output: Rendered Webpage: College Information Portal in Google Chrome

ANITS

VIZAG^{IN} Near Tagarapuvallasa

Yo, here's the tea on ANITS (Anil Neerukonda Institute of Technology and Sciences). It's an autonomous engineering college out in Sangivalasa, near Vizag, that popped up back in 2001. Going pro-teacher mode for a sec: the place flexes a shiny NAAC "A+" grade and is affiliated with Andhra University, heavily focusing on B.Tech programs like CSE, AI/ML, and Data Science. It sits on a 12-acre campus where students are basically grinding through assignments and labs trying to survive those placements. Lowkey, it's a solid spot if you're trying to secure the bag in the tech world without totally losing your mind.

1. Home

2. [Student Portal](#)

3. Alums

4. About us

• [Main Portal](#)

• [Teacher Portal](#)

• Alums

• About us

 **ANITS**

Anil Neerukonda Institute of Technology & Sciences (ANITS), Visakhapatnam is owned by Megha Engineering and Infrastructure Limited (MEIL) which is one of the top infrastructure and manufacturing company in India with headquarters at Hyderabad. The institute was established in the academic year 2001-02 by Anil Neerukonda Educational Society (ANES) which was founded by Dr. N.B.R. Prasad, an NRI Philanthropist from the USA, with industrialists and eminent educationists in memory of his son Late Anil Neerukonda.

ROLL	NAME	PLACED IN
A24126552259	J SATYA SRIKAR	GOOGLE
A24126552266	RSVR PRASAD	TESLA
A24126552268	V S SRAVAN	OPEN AI

The form

Name

Email

Password

Gender ☐ Male ☐ Female

Hobbies ☐ Playing Sport

☐ Singing

☐ Dancing

☐ Reading Books

☐ Free Fire

CSM color Range Number Date

▶ 0:00 / 0:00

THANK YOU

Figure 1: College Information Portal displaying header, placement records table, interactive registration form, and media controls.

10. RESULT

Thus, the webpage displaying College Information (incorporating fundamental elements, layouts blocks, grids database tables, custom styled forms, and multimedia players) was successfully designed, implemented, and verified.

Submission Package Structure:

Observation/Observation task 1/

■■■ Docs/

■ ■■■ ObsWeek1.docx

■ ■■■ ObsWeek1.pdf

■■■ Html/

■ ■■■ college-portal.html

■■■ Media/

■■■ Clg Logo.jpg

■■■ song.mp3